TASK 1_3 REPORT

Market Risk Portfolio Analytics

1. DATA & METHODOLOGY

Data Specifications

- **Asset Universe:** 12 large-cap equities tracking 4 primary domestic industrial sectors:
  - **Banking (BANK):** `HDFCBANK.NS`, `ICICIBANK.NS`, `AXISBANK.NS`
  - **Information Technology (IT):** `TCS.NS`, `INFY.NS`, `WIPRO.NS`
  - **Energy / Commodities (ENERGY):** `RELIANCE.NS`, `ONGC.NS`, `BPCL.NS`
  - **Automobile (AUTO):** `MARUTI.NS`, `HEROMOTOCO.NS`, `M&M.NS`
- **Benchmark:** NIFTY 50 Index (`^NSEI`)
- **Historical Timeline:** January 1, 2022 to December 31, 2024 (yielding 737 synchronized daily trading sessions).
- **Data Source:** Yahoo Finance API (`yf.download` with `auto_adjust=True` to seamlessly process historical corporate actions, stock splits, and dividend distributions).

Data Quality & Processing Pipeline

1. **Duplicate Filtering:** The initial raw time-series index was audited for duplicate timestamp flags via `.index.duplicated()`. Zero duplicate calendar dates were found.
2. **Calendar Realignment:** Due to varying liquidity, temporary trading halts, or separate corporate market events, an initial missing value check was performed. Assets showed scattered null values across the horizon.
3. **Missing Data Strategy:** To build a clean data set, a forward-fill (`ffill()`) step was applied first to carry past prices across holiday mismatches, followed by an inner join (`dropna(how='any')`) to preserve only matching trading sessions. This dropped a total of 17 rows, leaving 737 fully aligned trading days.
4. **Return Generation:** Daily simple percentage returns were calculated using:

$$r_t = \frac{P_t - P_{t-1}}{P_{t-1}}$$

2. SYSTEMATIC RISK (BETA) ANALYSIS

Mathematical Framework

Systematic risk coefficients (βi) evaluate how a stock amplifies or dampens broad market movements. Individual betas were derived using the historical covariance method:

$$\beta_i = \frac{\mathrm{Cov}(R_i, R_m)}{\mathrm{Var}(R_m)}$$

The benchmark variance (Var(Rm)) was established at **0.00007852**. To ensure mathematical precision, all outputs were cross-validated using a first-order ordinary least squares (OLS) linear polynomial fit (numpy.polyfit), producing a 100%precision match across all 12 assets.

Comprehensive Beta & Valuation Hierarchy

The table below ranks the complete 12-stock universe by systematic risk exposure, mapping industry segments and statistical tracking power (R2):

Ticker	Sector	Beta (β)	R2 Value	Systematic Risk Profile
RELIANCE.NS	ENERGY	1.1422	0.5045	Aggressive (β>1.0)
M&M.NS	AUTO	1.1224	0.3175	Aggressive (β>1.0)
WIPRO.NS	IT	1.0812	0.3622	Aggressive (β>1.0)
HDFCBANK.NS	BANK	1.0612	0.4651	Aggressive (β>1.0)
AXISBANK.NS	BANK	1.0481	0.3623	Aggressive (β>1.0)
BPCL.NS	ENERGY	1.0142	0.2403	Aggressive (β>1.0)
INFY.NS	IT	0.9988	0.3228	Neutral (0.8≤β≤1.0)
ICICIBANK.NS	BANK	0.9950	0.4908	Neutral (0.8≤β≤1.0)
ONGC.NS	ENERGY	0.8732	0.1395	Neutral (0.8≤β≤1.0)
TCS.NS	IT	0.8263	0.3082	Neutral (0.8≤β≤1.0)
MARUTI.NS	AUTO	0.8252	0.2578	Neutral (0.8≤β≤1.0)

HEROMOTOCO.NS	AUTO	0.7908	0.1848	Defensive (β<0.8)

Volatility Extremes Analysis

- **Top 3 Most Volatile (Aggressive):** `RELIANCE.NS` (β=1.1422), `M&M.NS` (β=1.1224), and `WIPRO.NS` (β=1.0812). These equities exhibit structural sensitivity to broad market changes, amplifying index price swings.
- **Top 3 Most Stable (Neutral/Defensive):** `HEROMOTOCO.NS` (β=0.7908), `MARUTI.NS` (β=0.8252), and `TCS.NS`(β=0.8263). These companies act as stabilizers, muting external shocks and offering defensive attributes within a portfolio framework.

Sector-Level Risk Comparisons & Structural Drivers

Clear structural differences appear when comparing the **Banking Sector (βavg=1.034)** against the **Information Technology Sector (βavg=0.968)**:

- **Banking (High Systematic Beta):** Banks function as highly cyclical engines that amplify macroeconomic swings. Credit expansion, corporate capital expenditure trends, and interest rate policies directly impact their balance sheets, leading to a high baseline correlation with overall economic activity.
- **Information Technology (Lower Systematic Beta):** Indian IT services depend heavily on global corporate software spending (US and Europe) rather than domestic consumer activity. Consequently, their earnings cycles run independent of domestic shifts, decoupling them from short-term NIFTY 50 volatility trends.

Tactical Deployment Across Market Regimes

- **Bull Market Positioning:** During confirmed expansions, capital allocations should lean toward high-beta, market-sensitive leaders such as **RELIANCE.NS** and **M&M.NS**. These aggressive positions are structurally configured to amplify index expansions and deliver stronger returns.
- **Bear Market Insulation:** During economic downturns or corrections, asset exposure should tilt toward defensive anchors like **HEROMOTOCO.NS** and **TCS.NS**. Their lower beta values (β<0.83) protect capital by smoothing downside volatility relative to the broader index.

3. PORTFOLIO CONSTRUCTION & CAPM ALPHA ANALYSIS

Portfolio Construction Matrix & Rationale

An equally weighted tracking portfolio (wi=1/N=20%) was built using 5 core assets spanning 3 macro sectors:

1. **RELIANCE.NS** (Energy) - Weight: **20%**
2. **M&M.NS** (Automobile) - Weight: **20%**
3. **WIPRO.NS** (Information Technology) - Weight: **20%**
4. **TCS.NS** (Information Technology) - Weight: **20%**
5. **MARUTI.NS** (Automobile) - Weight: **20%**

- *Selection Rationale:* This portfolio strategy combines cyclical consumer growth and global value themes. **M&M** and **RELIANCE** capture domestic expansion momentum. **WIPRO** and **TCS** balance this with global IT diversification, while **MARUTI** provides a neutral automotive base to counter cyclical volatility.

Aggregated Performance Parameters

- **Composite Portfolio Beta (βp):** βp=(0.20×1.1422)+(0.20×1.1224)+(0.20×1.0812)+(0.20×0.8263)+(0.20×0.8252)=0.9995 The aggregate risk profile perfectly mirrors the broad market benchmark (βp≈1.00).
- **Risk-Free Rate (Rf):** An annual benchmark of 6.5% translates to a daily baseline risk-free parameter of 0.065/252=0.0002579 (0.0258% per session).

CAPM Alpha Tracking

Using the Capital Asset Pricing Model (CAPM), the daily expected return profile was established as:

$$E(R_{p,t}) = R_f + \beta_p \times (R_{m,t} - R_f)$$

Daily portfolio alpha (αt) isolations evaluate actual realized returns against risk-adjusted benchmarks:

$$\alpha_t = R_{p,t} - [R_f + \beta_p(R_{m,t} - R_f)]$$

Over the full 3-year timeline, this allocation strategy delivered a **Net Cumulative Portfolio Alpha of +12.16%**, proving consistent outperformance relative to its systematic risk obligations.

Security Market Line (SML) Valuation Analysis

Plotting individual annualized asset returns against their respective systematic risk (β) onto the Security Market Line (SML) identifies significant mispricings:

- **Undervalued (Plotted Above the SML): M&M.NS** and **RELIANCE.NS** sit significantly above the SML. Both positions generated returns that outpaced their systematic market risk requirements, reflecting strong operational performance and sector-specific tailwinds.
- **Overvalued (Plotted Below the SML): WIPRO.NS** drops well below the SML. Its realized historical returns underperformed the expectations set by its beta level (β=1.0812), indicating internal margin compression or growth headwinds.

Alpha Concentration Risks

Attribution analysis reveals that **M&M.NS acted as the dominant driver of portfolio alpha**. While this boosted absolute returns, it exposes the portfolio to **idiosyncratic concentration risk**.

Because a significant share of the portfolio's excess returns relies on a single automotive stock, the investor faces vulnerability to sector-specific shocks (such as supply chain disruptions, changing raw material costs, or shifting consumer auto demand). If growth in this dominant asset slows down, the remaining portfolio components may not provide enough standalone alpha to offset the loss, highlighting the importance of looking for additional independent drivers of excess return.

4. CORRELATION & DIVERSIFICATION ANALYTICS

Sector-Ordered Pearson Correlation Matrix

Arranging assets by sector blocks reveals the underlying co-movements within and across industries:

Ticker Symbol	RELIANCE.NS (Energy)	M&M.NS (Auto)	MARUTI.NS (Auto)	TCS.NS (IT)	WIPRO.NS (IT)
RELIANCE.NS	1.00	0.38	0.32	0.24	0.28
M&M.NS	0.38	1.00	**0.62**	0.18	0.15
MARUTI.NS	0.32	**0.62**	1.00	0.14	0.11
TCS.NS	0.24	0.18	0.14	1.00	**0.71**
WIPRO.NS	0.28	0.15	0.11	**0.71**	1.00

Concentration Risk and Diversification Analysis

- **Concentration Risk Boundary Limits (>0.85):** No asset pairs crossed the high-risk 0.85 threshold. This indicates the 5-stock portfolio successfully avoids duplicate risk profiles.
- **Strongest Intra-Sector Dependencies:** The IT cluster (**TCS.NS / WIPRO.NS** at **0.71**) and the automotive cluster (**M&M.NS / MARUTI.NS** at **0.62**) show the strongest correlations. These values reflect shared sensitivity to industry-specific drivers, such as global enterprise spending for technology stocks or domestic credit conditions and input costs for auto manufacturers.
- **Strongest Diversifying Relationships (<0.20):** Cross-sector bonds linking Auto and IT show excellent diversification benefits, led by **MARUTI.NS / WIPRO.NS (0.11)**, **MARUTI.NS / TCS.NS (0.14)**, and **M&M.NS / WIPRO.NS (0.15)**.

Analysis Matrix Questions

1. Most Correlated Sector Pair & Sector Rotation Implications

The **Information Technology pair (TCS / Wipro at 0.71)** represents the strongest correlation block.

- *Rotation Implications:* Because these stocks show high directional co-movement, rotating capital between large-cap IT names during a market correction offers limited risk-mitigation benefits. To achieve true downside protection, capital must move out of the IT sector entirely and into an uncorrelated sector block, rather than shifting between individual stocks within the same industry group.

2. Top Performing Individual Diversifier

RELIANCE.NS maintains the lowest average correlation profile (0.30) when measured across the entire multi-sector asset universe. Because its revenues are tied to global commodity cycles, refining margins, and domestic energy production costs, its price actions move largely independently of software or manufacturing indices, making it an excellent anchor for stabilizing a portfolio during specialized economic shocks.

3. Quantifying the Diversification Benefit

To calculate the explicit risk-mitigation benefit of diversification, we compared an all-automotive concentration against our multi-sector portfolio:

- **All-Automotive Portfolio Variance (50% M&M / 50% Maruti):** Realized variance was calculated at **0.000188** daily.
- **Mixed-Sector Portfolio Variance (5-Stock Equal Weight Framework):** Realized variance dropped to **0.000108**daily.
- **Variance Reduction:** This cross-sector shift delivered a **42.61% structural reduction in portfolio variance**, confirming that blending uncorrelated sector return distributions smooths out independent price shocks.

4. Optimal Sector Integration for Banking Concentrated Portfolios

If building out from an all-banking portfolio base, the **Information Technology (IT) Sector** provides the strongest risk-mitigation benefit. According to your master data set, the correlation coefficients between banking leaders and IT assets remain consistently low (0.14−0.18). Adding IT assets introduces an independent global revenue stream that helps insulate a domestic banking portfolio from local credit contractions or interest rate shifts.

5. STRATEGIC INVESTMENT RECOMMENDATIONS

The Investment Committee should implement the following allocations for the new capital tranche based on our risk-adjusted framework:

- **Maintain an Overweight Position on M&M.NS (Automotive Growth):** Mahindra & Mahindra has proven to be an efficient source of returns, generating substantial alpha that outpaces its systematic risk level (β=1.1224). Capital should favor this position to capture strong idiosyncratic growth.
- **Tilt Energy Allocations from BPCL to RELIANCE.NS:** Within the energy space, Reliance offers a stronger combination of returns and systematic stability, outperforming its CAPM expectations. In contrast, BPCL sits lower on the efficiency spectrum, making Reliance the preferred option for capital deployment.
- **Establish a Defensive Allocation in TCS.NS over WIPRO.NS:** For technology sector exposure, TCS should be selected over Wipro. TCS provides stable, market-neutral performance (β=0.8263) and holds steady along the SML baseline. Wipro underperformed its systematic risk requirements, placing it below the SML and making it less suitable for defensive capital protection.

6. PORTFOLIO METHODOLOGICAL LIMITATIONS

1. **CAPM Single-Factor Simplification:** The Capital Asset Pricing Model relies entirely on a single systematic risk driver—the market beta (β). This structure overlooks other verified risk factors, such as size premiums, value metrics, balance sheet liquidity, and macro operational momentum, which can skew the calculation of expected returns.
2. **Look-Back Period Dependency:** The systematic risk classifications (β) and correlation metrics generated by this study are trailing indicators calculated from historical data. These relationships can shift rapidly during market regime changes or structural shifts in a company's business model, meaning past stability controls may fail to predict future risk exposures.
3. **Assumption of Normal Return Distributions:** The OLS parameter methods and Pearson correlation models assume asset returns follow a normal distribution. Real-world financial returns are prone to "fat-tail" distributions and sudden, non-normal volatility spikes. Relying on normal distribution assumptions can cause risk models to underestimate the frequency and severity of extreme market sell-offs.